

Prac-03

Create and run a MySQL Docker container using MySQL 8.x.

Access the running container using docker exec as an admin user, display all databases, and switch to mydatabase.

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\VICTUS> mkdir dockerfile-prac2

Directory: C:\Users\VICTUS

Mode                LastWriteTime         Length Name
----                -
d-----          03-05-2026   01:45 PM             dockerfile-prac2

PS C:\Users\VICTUS> cd dockerfile-prac2
PS C:\Users\VICTUS\dockerfile-prac2> notepad dockerfile
PS C:\Users\VICTUS\dockerfile-prac2> ren dockerfile.txt dockerfile
PS C:\Users\VICTUS\dockerfile-prac2>
```

Step1: create the file

Step2: get inside the file and add the content

FROM mysql:8.1

ENV MYSQL_ROOT_PASSWORD=root123

ENV MYSQL_DATABASE=mydatabase

ENV MYSQL_USER=admin

ENV MYSQL_PASSWORD=root123

EXPOSE 3306

Step3: change the dockerfile.txt

```
Terminal

PS C:\Users\VICTUS\dockerfile-prac2> docker build -t mysql:8.1 .
[+] Building 19.3s (6/6) FINISHED                                docker:desktop-linux
=> [internal] load build definition from dockerfile              0.1s
=> => transferring dockerfile: 182B                               0.0s
=> [internal] load metadata for docker.io/library/mysql:8.1     3.4s
=> [auth] library/mysql:pull token for registry-1.docker.io     0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [1/1] FROM docker.io/library/mysql:8.1@sha256:f61944ff3f2961363a4d22913b2 8.0s
=> => resolve docker.io/library/mysql:8.1@sha256:f61944ff3f2961363a4d22913b2 0.0s
=> => sha256:8e0176adc18c476bdfcc701f01cda5acf49bc8e6d7fad 44.28MB / 44.28MB 3.8s
=> => sha256:fba70cc872a5b5a727ef28a9ed8203ad526cb069dc21dca 4.61MB / 4.61MB 2.6s
=> => sha256:47f08b0e916e6e60dc941429579e860a2f1c4b6eeadd5600ba 317B / 317B 1.5s
=> => sha256:850e29ae8eeb4a7f347575c2fa8f159be328eacac69cc 57.48MB / 57.48MB 5.3s
=> => sha256:86bf2dc4ded9433a1f0b2155401a02843311dc129425 58.75MB / 58.75MB 5.4s
=> => sha256:a7b58dd6f78b98d9ff6fbc049f2f7cb84174c74dd57 982.81kB / 982.81kB 1.1s
=> => sha256:081f41f573ba759e534574da506eb0a6593adb408cf88d3f9c5 339B / 339B 0.8s
=> => sha256:5db2cc6eab8f76e59babeec88038ba1e7d20bbdca06ae1 2.61kB / 2.61kB 0.7s
=> => exporting to image                                           15.7s
=> => exporting layers                                             0.0s
```

Step4: Build the image

```
Terminal
PS C:\Users\VICTUS\dockerfile-prac2> docker run -d -p 3306:3306 --name mysql-prac mysql:8.1
199f240b52186d6d98ed843a55f27cc53a13f78b2a25462b1817870305f78f68
PS C:\Users\VICTUS\dockerfile-prac2> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
199f240b5218   mysql:8.1     "docker-entrypoint.s..." 26 seconds ago Up 25 seconds 0.0.0.0:3306->3306/tcp, [::]:3306->33
c30807bc142f   kavya4619m/ab2:v1.0 "/bin/bash"             51 minutes ago Up 51 minutes
PS C:\Users\VICTUS\dockerfile-prac2> docker exec -it mysql-prac mysql -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.1.0 MySQL Community Server - GPL

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```

Step5: Run the container

```
Terminal

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mydatabase |
| performance_schema |
+-----+
3 rows in set (0.00 sec)

mysql> USE mydatabase;
Database changed
mysql>
```

RAM 3.50 GB CPU 0.08% Disk: 21.12 GB used (limit 1006.85 GB)

Step6: get inside the container

Step7: show all the database and switch to mydatabase